A Connes–Dixmier-measurable operator that is not Dixmier measurable

Candidate counterexample for arXiv:math/0608375

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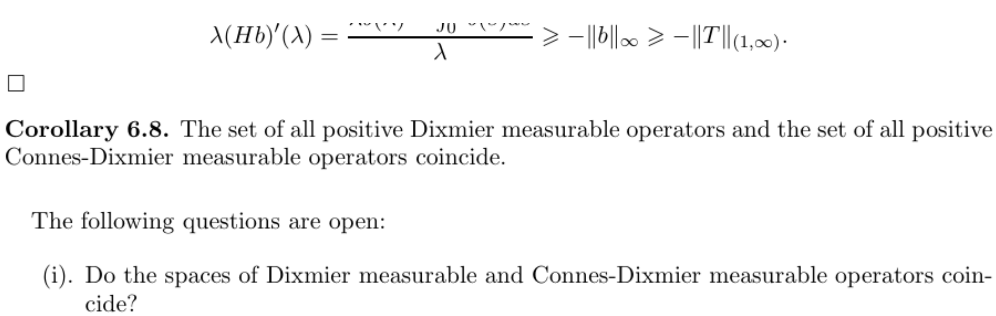
Status. Candidate full negative answer to Question (i) of Carey–Sukochev, subject to expert review.

1 Source question and result

Carey and Sukochev ask, immediately after Corollary 6.8 on page 22 of their survey [1], whether the spaces of Dixmier-measurable and Connes–Dixmier-measurable operators coincide. Their corollary proves coincidence on the positive cone of the Dixmier–Macaev ideal $\mathcal{L}^{(1,\infty)}$.



$$\lambda(Hb)'(\lambda) = \frac{\cdots}{\lambda} \geqslant -\|b\|_\infty \geqslant -\|T\|_{(1,\infty)}.$$

□

Corollary 6.8. The set of all positive Dixmier measurable operators and the set of all positive Connes-Dixmier measurable operators coincide.

The following questions are open:

(i). Do the spaces of Dixmier measurable and Connes-Dixmier measurable operators coincide?

Figure 1: The open question in [1, p. 22].

We answer the question negatively, already for a diagonal self-adjoint operator on $\ell_2 \oplus \ell_2$.

For a positive compact operator T, put

$$a_T(t) = \frac{1}{\log(1+t)} \int_0^t \mu_s(T)\, ds, \qquad t > 0.$$

If $T = T_+ - T_-$ is self-adjoint, its trace profile is

$$f_T(t) = a_{T_+}(t) - a_{T_-}(t).$$

Every Dixmier trace has the form $\mathrm{Tr}_\omega(T) = \omega(f_T)$ for a dilation-invariant generalized limit ω. A Connes–Dixmier trace uses $\omega = \gamma \circ M$, where

$$(Mf)(t) = \frac{1}{\log t} \int_1^t f(s)\, \frac{ds}{s}, \qquad t > 1,$$

and γ is a singular state. These are precisely the definitions used in [1, Sections 5–6].

Theorem 1. *There is a compact self-adjoint operator $B \in \mathcal{L}^{(1,\infty)}(\ell_2 \oplus \ell_2)$ such that*

1. $\mathrm{Tr}_\omega(B) = 0$ *for every Connes–Dixmier trace;*
2. *there are Dixmier traces* $\mathrm{Tr}_{\omega_0}, \mathrm{Tr}_{\omega_1}$ *satisfying*

$$\mathrm{Tr}_{\omega_0}(B) = 0, \qquad \mathrm{Tr}_{\omega_1}(B) = \frac{2}{\log 2}.$$

Consequently, the Connes–Dixmier-measurable and Dixmier-measurable operator spaces in $\mathcal{L}^{(1,\infty)}$ do not coincide.

2 Proof intuition

On logarithmic scale, the operator M is the ordinary Cesàro average. Thus a bounded profile made of longer and longer plateaux can still satisfy $Mf \to 0$ if those plateaux have asymptotic density zero. Every Connes–Dixmier trace then sees zero. On the other hand, an invariant mean can be obtained by averaging only over the plateaux; it sees their nonzero height. A second invariant mean can average over the much longer zero gaps.

The only real constraint is to realize this abstract profile by an operator. In the dyadic rank block $[2^n, 2^{n+1})$, prescribe total singular mass p_n. The singular values are decreasing exactly when $p_{n+1} \le 2p_n$. A doubling ramp therefore creates a macroscopic mass jump without violating monotonicity. A matching ramp in the negative Jordan part cancels the jump exactly after a long plateau. Superexponentially separated pulses keep both cumulative singular masses $O(n)$ through dyadic time n, which is exactly the $\mathcal{L}^{(1,\infty)}$ bound.

3 Construction

For $j \ge 2$, define

$$N_j = 2^{2^j}, \qquad L_j = 2^j, \qquad W_j = 2^{2^{j-1}} = \sqrt{N_j}.$$

In particular, $2^{L_j} = N_j$ and $N_{j+1} = N_j^2$. The intervals below are pairwise disjoint. Define two sequences of dyadic block masses by

$$p_n = \begin{cases} 2^r, & n = N_j - L_j + r \text{ for some } j \ge 2, 1 \le r \le L_j, \\ 1, & \text{otherwise,} \end{cases}$$

and

$$q_n = \begin{cases} 2^r, & n = N_j + W_j + r \text{ for some } j \ge 2, 1 \le r \le L_j, \\ 1, & \text{otherwise.} \end{cases}$$

For integer ranks k in the dyadic block $I_n = \{2^n, \dots, 2^{n+1} - 1\}$, set

$$u_k = \frac{p_n}{2^n}, \qquad v_k = \frac{q_n}{2^n}.$$

Assign any harmless common value at rank zero. Let

$$U = \mathrm{diag}(u_k), \qquad V = \mathrm{diag}(v_k), \qquad B = U \oplus (-V).$$

Lemma 1. *The sequences (u_k) and (v_k) are decreasing to zero, and $U, V \in \mathcal{L}^{(1,\infty)}$. Consequently $B \in \mathcal{L}^{(1,\infty)}$ and its Jordan parts are $B_+ = U \oplus 0$ and $B_- = 0 \oplus V$.*

Proof. The only upward changes in either block-mass sequence are the steps $1, 2, 4, \ldots, 2^{L_j}$. Hence always $p_{n+1} \le 2p_n$ and $q_{n+1} \le 2q_n$. Therefore

$$\frac{p_{n+1}}{2^{n+1}} \le \frac{p_n}{2^n}, \qquad \frac{q_{n+1}}{2^{n+1}} \le \frac{q_n}{2^n}.$$

The ratios tend to zero because every exceptional mass near logarithmic time N_j is at most N_j, whereas the block length is 2^n.

The excess mass in one complete ramp is

$$A_j := \sum_{r=1}^{L_j} (2^r - 1) = 2^{L_j+1} - 2 - L_j = 2N_j - 2 - L_j < 2N_j. \tag{1}$$

For any m, the sum $\sum_{n \le m} p_n$ is its baseline $m+1$ plus complete earlier ramp excesses and at most one current partial excess. If the jth ramp has begun, then $m \ge N_j - L_j + 1 \ge N_j/2$. Moreover, since $N_{i+1} = N_i^2$, $\sum_{i<j} N_i \le N_j$ (enlarging the harmless constant for the first few indices). Thus

$$\sum_{n=0}^{m} p_n \le C(m+1)$$

with an absolute C. The same argument applies to (q_n), whose ramp is even later than N_j. At the end of block I_m, cumulative singular mass is therefore $O(m)$, while $\log(1 + 2^{m+1}) \asymp m + 1$. The estimate inside a block follows by monotonicity. This is exactly the $\mathcal{L}^{(1,\infty)}$ bound. $\square$

4 Trace-profile analysis

Let $F = f_B$. The common baseline block masses cancel. During the positive ramp of pulse j, the numerator of F increases from zero to A_j; it is constant through the following W_j baseline blocks; the negative ramp has the same excesses in the same order and decreases it exactly back to zero. It follows that, in logarithmic coordinates,

$$\operatorname{supp}\bigl(v \mapsto F(2^v)\bigr) \subseteq \bigcup_{j \ge 2} [N_j - L_j + 1,\ N_j + W_j + L_j + 1]. \tag{2}$$

The profile is uniformly bounded: on the jth interval its numerator is between 0 and $A_j < 2N_j$, and its logarithmic denominator is comparable with N_j.

Lemma 2. *$MF(t) \to 0$ as $t \to \infty$.*

Proof. For $x > 0$, the change of variables $s = 2^v$ gives

$$(MF)(2^x) = \frac{1}{x} \int_0^x F(2^v)\, dv.$$

By (2), the logarithmic support accumulated up through a point of order N_j is at most

$$\sum_{i \le j} (W_i + 2L_i + 2) = O(W_j) = O(\sqrt{N_j}) = o(N_j).$$

The same estimate, with the last completed pulse, applies in every gap. Since F is uniformly bounded, its average on $[0, x]$ tends to zero. $\square$

Every Connes–Dixmier trace on B equals $\gamma(MF)$ for a singular state γ. Lemma 2 proves part (1) of Theorem 1.

For the converse separation, consider the plateau intervals

$$J_j = [N_j + 1, N_j + W_j + 1].$$

On J_j the numerator of $F(2^v)$ is exactly A_j, so uniformly for $v \in J_j$,

$$F(2^v) = \frac{A_j}{\log(1+2^v)} \longrightarrow \frac{2}{\log 2}, \tag{3}$$

because $A_j/N_j \to 2$ and $W_j/N_j \to 0$. The interval lengths $|J_j| = W_j$ tend to infinity.

Let $\mathcal{U}$ be a free ultrafilter and, for $g \in L_\infty(\mathbb{R}_+)$, define

$$\Lambda_1(g) = \lim_{j\to\mathcal{U}} \frac{1}{|J_j|} \int_{J_j} g(v)\, dv.$$

This is a generalized limit. It is translation invariant: translating a fixed distance changes an interval average by at most the norm of g times the relative boundary length, which tends to zero. Hence the functional

$$\omega_1(h) := \Lambda_1\big(v \mapsto h(2^v)\big)$$

is a dilation-invariant generalized limit on $L_\infty(\mathbb{R}_+^*)$. Equation (3) gives $\omega_1(F) = 2/\log 2$.

After the negative ramp, F is identically zero until the next positive ramp. In particular, for all sufficiently large j it vanishes on

$$K_j = [2N_j, 3N_j],$$

because the jth pulse ends before $2N_j$ and the next begins near $N_{j+1} = N_j^2$. Repeating the interval-ultralimit construction with K_j gives a dilation-invariant generalized limit ω_0 satisfying $\omega_0(F) = 0$. The corresponding Dixmier traces disagree on B. This proves part (2) and Theorem 1.

5 Verification and novelty boundary

The script `code/verify_pulses.py` checks, for increasingly enormous exact integer parameters, the ramp identity (1), exact positive– negative cancellation, the local inequalities $p_{n+1} \le 2p_n$ and $q_{n+1} \le 2q_n$, the prefix-growth estimate, and convergence of the plateau height to $2/\log 2$.

A bounded novelty search used the exact source wording and the phrases "Connes–Dixmier measurable but not Dixmier measurable", "non-positive", "self-adjoint", and "$\mathcal{M}_{1,\infty}$". Sukochev–Usachev–Zanin [2] prove that the two trace classes differ on $\mathcal{M}_{1,\infty}$, but their positive measurable-operator separation uses a different Marcinkiewicz ideal; their paper reiterates positive coincidence on $\mathcal{M}_{1,\infty}$. Semenov–Sukochev–Usachev–Zanin [3, Theorem 7.7] later drop positivity on the strictly smaller weak trace ideal $\mathcal{L}_{1,\infty}$ and explicitly label that scope. No source found in this search states the arbitrary-operator separation above in the original $\mathcal{L}^{(1,\infty)}$.

The result addresses only Question (i) of [1]; it gives no description for the maximally invariant or all-symmetric-functional classes in Questions (ii) and (iii).

6 Human-review recommendation

Review as a high-priority candidate full counterexample. The most useful checks are: (a) the uniform prefix bound in Lemma 1; (b) the support-density estimate in Lemma 2; and (c) that the interval ultralimits meet exactly the source definition of dilation-invariant states.

References

[1] A. L. Carey and F. A. Sukochev, *Dixmier traces and some applications to noncommutative geometry*, Russian Math. Surveys 61 (2006), 1039–1099; arXiv:math/0608375.

[2] F. Sukochev, A. Usachev, and D. Zanin, *On the distinction between the classes of Dixmier and Connes–Dixmier traces*, Proc. Amer. Math. Soc. 141 (2013), 2169–2179; arXiv:1210.3397.

[3] E. Semenov, F. Sukochev, A. Usachev, and D. Zanin, *Banach limits and traces on $\mathcal{L}_{1,\infty}$*, Adv. Math. 285 (2015), 568–628; arXiv:1612.04509.